

SDM in Shiny

SDM in Web based platforms

1. Intro **Shiny**-Why ?
2. How it works in SDM?

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What is Shiny (R)?

- Shiny is an R package that enables building interactive web applications that can execute R code on the backend;
- With Shiny, you can host standalone applications on a webpage, embed interactive charts in R Markdown documents, or build dashboards;

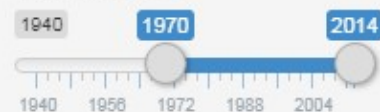
Movie explorer

Filter

Minimum number of reviews on Rotten Tomatoes



Year released



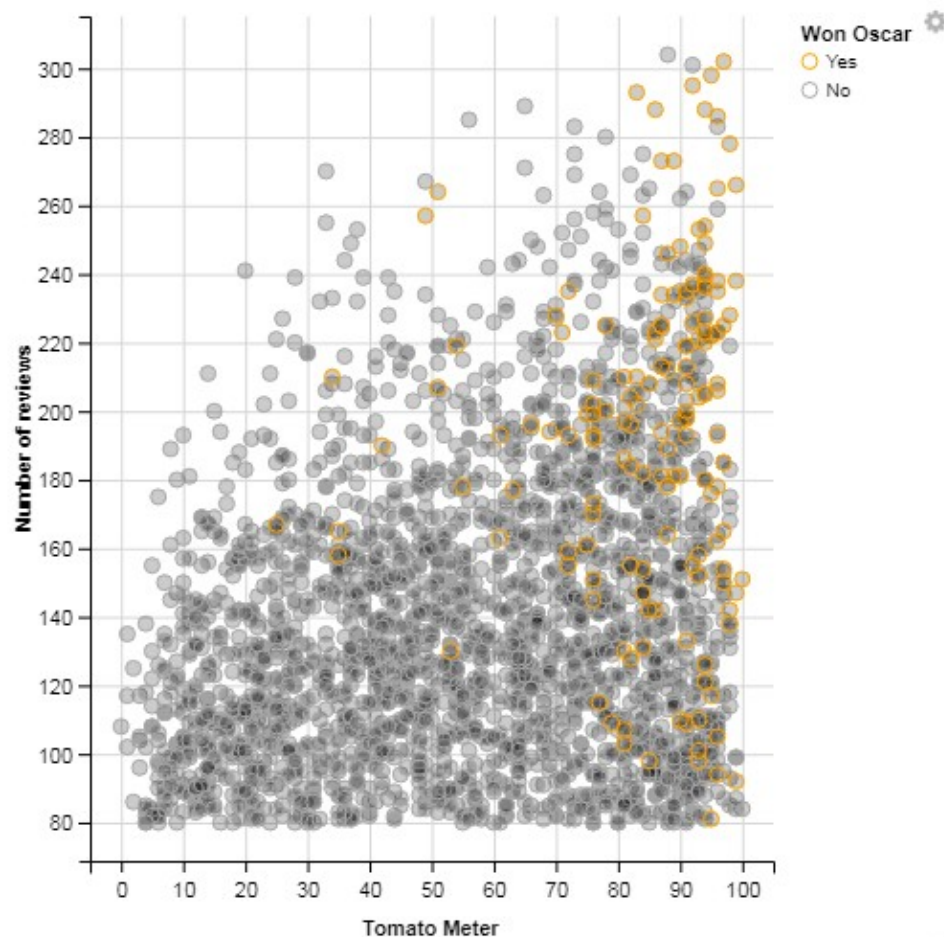
Minimum number of Oscar wins (all categories)



Dollars at Box Office (millions)



Genre (a movie can have multiple genres)



Number of movies selected:
2557

Web based platform in SDM

- Wallace: very successful, widely used for conservation training

```
install.packages("wallace")
```

```
library(wallace)
```

```
run_wallace()
```

- ntbox

```
run_ntbox
```

- shinybiomod

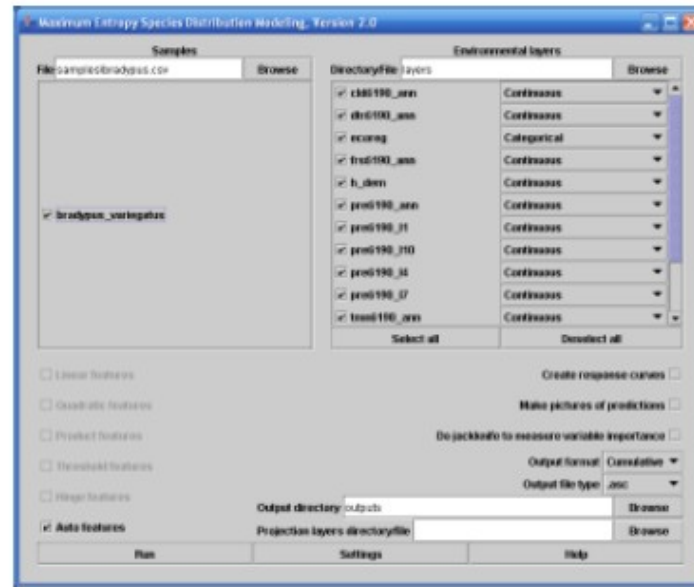
Biomod2 is very successful but ShinyBiomod is still young

<https://gitlab.com/lanOndo/shinybiomod>

Idea: run R code with GUI workflows

Foreground:

Extensible, flexible,
informative GUI

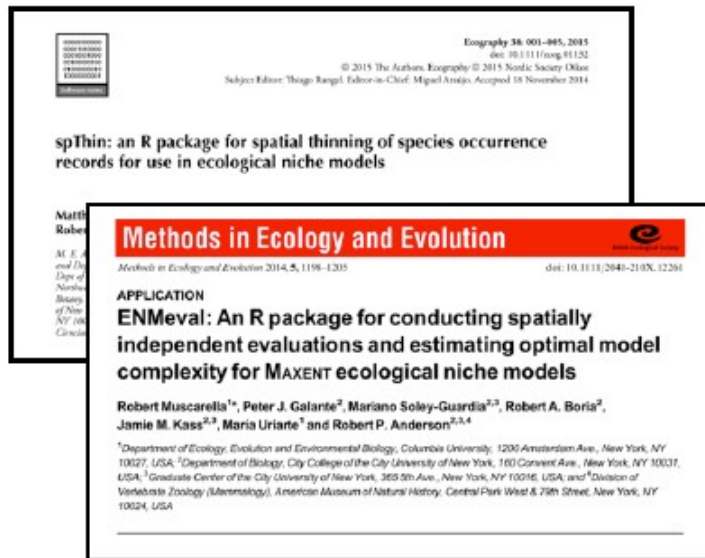


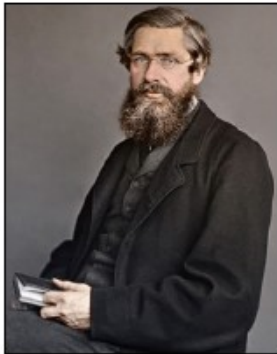
Background:

R code from new
packages

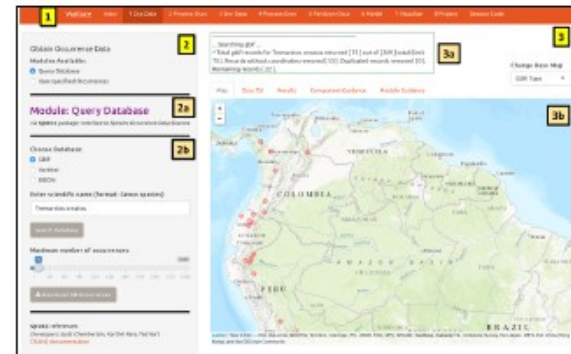
```
33 results <- occ(query = "Tremarctos ornatus", from = "gbif", limit = 81, has_coords = TRUE)
34 results.data <- results[["gbif"]]$data[[formatSpName("Tremarctos ornatus")]]
35 occs <- remDups(results.data) # remove rows with duplicate coordinates
36 ~~~
37
38 ~~~ Process Occurrence Data
39
40 You chose 42 of 51 total occurrence localities via polygon selection to keep in the analysis.
41
42 ~~~[r]
43 occs <- occs[c(1, 2, 3, 5, 6, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 20, 21, 22, 24, 25, 26, 27,
44 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51), ] # subset occs by selected rows
```

Idea: feature code-based methods with shiny GUI





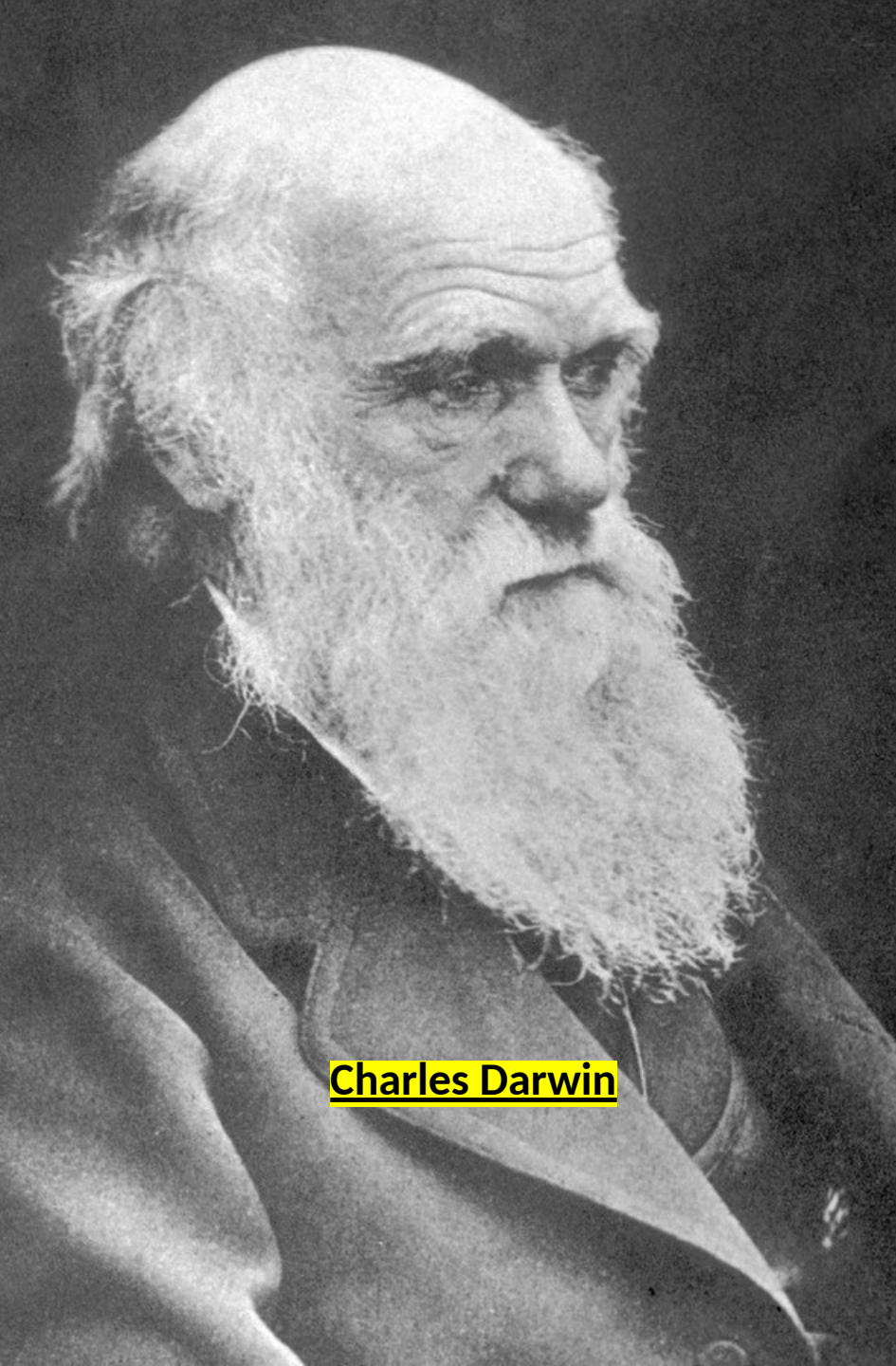
Wallace



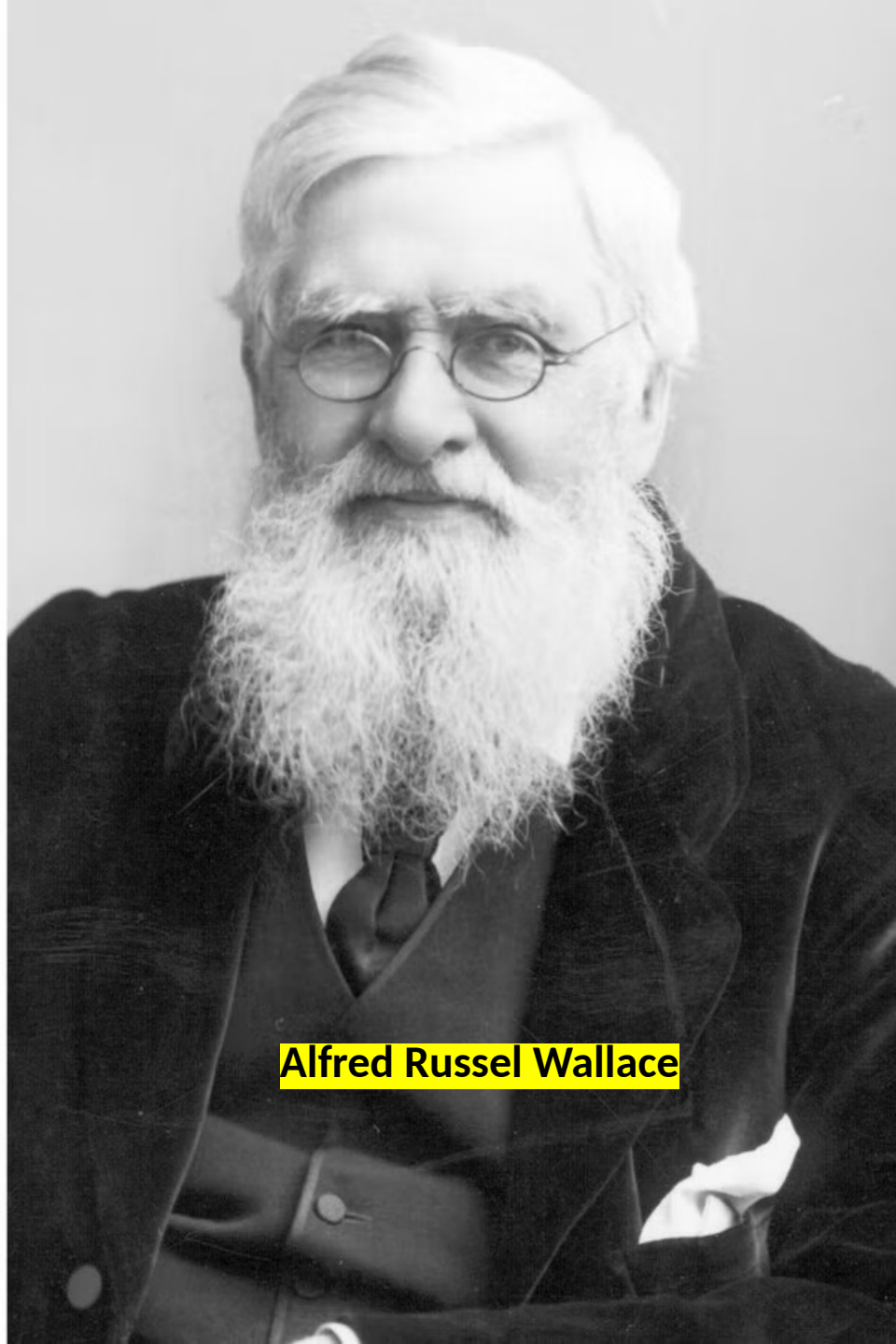
What is *Wallace*? And Why?

Point-and-click (GUI) application that features a niche/distribution modeling workflow

<https://wallacecomod.github.io/>



Charles Darwin



Alfred Russel Wallace

Wallace ecological modeling application

Qualities:

open
instructive
flexible
interactive
reproducible
expandable



1 Wallace Intro 1 Occ Data 2 Process Occs 3 Env Data 4 Process Envs 5 Partition Occs 6 Model 7 Visualize 8 Project Session Code

2 Obtain Occurrence Data

Modules Available:

- ☒ Query Database
- ☐ User-specified

2a Module: Query Database

SPDCC: interface to Species Occurrence Data Sources

Choose Database

- ☒ GBIF
- ☐ VertNet
- ☐ BiSON

Enter species scientific name

Tremarctos ornatus

Set maximum number of occurrences

75

Query Database

Download database occurrence localities (.csv)

Download

Module Developers: Jamie M. Kass, Bruno Vilela, Robert P. Anderson

SPDCC references

Package Developers: Scott Chamberlain, Karthik Ram, Ted Hart

CRAN | documentation

Please find messages for the user in this log window.

3a

3 Change Base Map

ESRI Topo

3b

Map Occs Tbl Results Component Guidance Module Guidance

A map of the Amazon Basin region, showing countries like Venezuela, Colombia, Ecuador, Peru, and Brazil. Red dots indicate occurrence data points for Tremarctos ornatus. The map includes a scale bar and a legend.

APPLICATION

WALLACE: A flexible platform for reproducible modeling of species niches and distributions built for community expansion

Jamie M. Kass^{1,2} | Bruno Vilela³ | Matthew E. Aiello-Lammens⁴ |
Robert Muscarella⁵ | Cory Merow⁶ | Robert P. Anderson^{1,2,7}

Methods in Ecology and Evolution



<https://wallaceecomod.github.io/>

ntbox

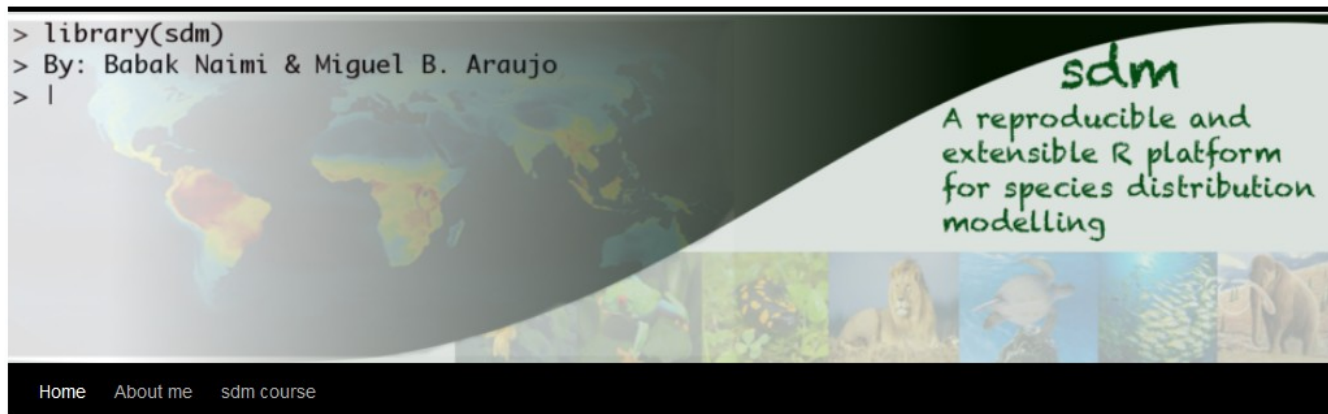
[library\(ntbox\)](#)
[run_ntbox\(\)](#)

1. **AppSettings** configuring all you need
2. The **Data** section
3. **Niche space**
4. **Niche correlations**
5. **Niche clustering**
6. **ENM** (Ecological niche modeling)
7. **SDM performance**
8. **Extrapolation risk** (model uncertainty)
9. **GIS tools**
10. **Saving your analysis**

- https://luismurao.github.io/ntbox_user_guide.html

sdm package

<https://www.biogeoinformatics.org;>



sdm

getting started with sdm package

[Babak Naimi](#), [Miguel B. Araujo](#)

3 April 2016 (update: 17 Oct. 2016)

<https://sdm.r-forge.r-project.org/>

- Simplized version of biomod2;
- sdm is an object-oriented, reproducible and extensible, platform for species distribution modelling.
- Developed by Babak Naimi and **Miguel B. Araújo**

- `library(sdm)`
- `file <- system.file("external/model.sdm",
package="sdm")`
- `m <- read.sdm(file) # a sdmModels Object (fitted using
sdm function)`
- `m`
- `gui(m)`

What is the limitation?

It is very useful for beginners and training, but has limitations:

- (1) Many functions can not work out in the platform ;
 - (2) Some specific/personalized requirements still have to be work out in R code, like repetition running (loop)
 - (3) limited capacity on computing, running big data
- ...